

HDOC DAY-26

Title: Right-Aligned Pattern Generation using Parameterless Functions and Nested Loops

1. Problem Statement Write a menu-driven C program using a switch statement to print various "Right-aligned patterns" selected by the user. Use nested loops without arrays. Each pattern activity must be implemented in a separate parameterless function returning void (void function_name(void)):

1. **Right-aligned increasing triangle:** Input n and print a right-aligned increasing triangle using asterisks.
2. **Right-aligned increasing number triangle:** Input n and print natural numbers from 1 up to the corresponding row number in a right-aligned format.
3. **Right-aligned decreasing triangle:** Input n and print a right-aligned decreasing triangle using asterisks.
4. **Exit:** Terminate program loop.

2. Algorithms

Main Control Flow

1. Display Right-Aligned Patterns Menu inside a do-while loop.

2. Read choice and route through switch statement to execute corresponding parameterless function.
3. Exit when choice 4 is entered.

Pattern Functions

- **print_right_aligned_increasing_triangle():** Read n . Loop i from 1 to n : print $n - i$ spaces, then inner loop j from 1 to i : print $*$.
- **print_right_aligned_increasing_number_triangle():** Read n . Loop i from 1 to n : print $n - i$ spaces, then inner loop j from 1 to i : print j .
- **print_right_aligned_decreasing_triangle():** Read n . Loop i from n down to 1: print $n - i$ spaces, then inner loop j from 1 to i : print $*$.

3. Test Case Table

Test Case	Pattern / Option	Input n	Expected Pattern Output	Actual Output	Status
1	Right-Aligned Increasing Triangle	$n = 4$	4 right-aligned star rows (1,2,3,4)	Matched expected output	Pass

Test Case	Pattern / Option	Input n	Expected Pattern Output	Actual Output	Status
2	Right-Aligned Increasing Numbers	$n = 4$	4 right-aligned number rows (1/12/123/1234)	Matched expected output	Pass
3	Right-Aligned Decreasing Triangle	$n = 4$	4 right-aligned star rows (4,3,2,1)	Matched expected output	Pass
4	Invalid Choice	Choice = 8	Error message displayed	Error message shown	Pass
5	Exit Option	Choice = 4	"Exiting program ..."	Program exited	Pass

4. Source Code Listing

```
#include <stdio.h>

void print_right_aligned_increasing_triangle(void) {
    int n;
    printf("\n--- Right-Aligned Increasing Triangle ---\n");
    printf("Enter number of rows (n): ");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Input must be a positive integer.\n"); return; }
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) printf(" ");
        for (int j = 1; j <= i; j++) printf("* ");
        printf("\n");
    }
}

void print_right_aligned_increasing_number_triangle(void) {
    int n;
    printf("\n--- Right-Aligned Increasing Number Triangle ---\n");
    printf("Enter number of rows (n): ");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Input must be a positive integer.\n"); return; }
    for (int i = 1; i <= n; i++) {
        for (int s = 1; s <= n - i; s++) printf(" ");
        for (int j = 1; j <= i; j++) printf("%d ", j);
        printf("\n");
    }
}

void print_right_aligned_decreasing_triangle(void) {
    int n;
    printf("\n--- Right-Aligned Decreasing Triangle ---\n");
    printf("Enter number of rows (n): ");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Input must be a positive integer.\n"); return; }
    for (int i = n; i >= 1; i--) {
        for (int s = 1; s <= n - i; s++) printf(" ");
        for (int j = 1; j <= i; j++) printf("* ");
        printf("\n");
    }
}
```

```

    }
}

int main(void) {
    int choice;
    do {
        printf("\n===== RIGHT-ALIGNED PATTERNS MENU
=====\\n");
        printf("1. Right-aligned increasing triangle\\n");
        printf("2. Right-aligned increasing number triangle\\n");
        printf("3. Right-aligned decreasing triangle\\n");
        printf("4. Exit\\n");
        printf("Enter choice (1-4): ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: print_right_aligned_increasing_triangle(); break;
            case 2: print_right_aligned_increasing_number_triangle(); break;
            case 3: print_right_aligned_decreasing_triangle(); break;
            case 4: printf("Exiting program...\\n"); break;
            default: printf("Error: Invalid choice!\\n"); break;
        }
    } while (choice != 4);
    return 0;
}

```

4. Execution & Verification Screenshots

```
kali@kali: ~/HDOOC_Day26
Session Actions Edit View Help
(kali@kali)~$ mkdir HDOOC_Day26
(kali@kali)~/HDOOC_Day26$ nano day26.c
(kali@kali)~/HDOOC_Day26$ gcc day26.c -o day26
(kali@kali)~/HDOOC_Day26$ ./day26

===== RIGHT-ALIGNED PATTERNS MENU =====
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter choice (1-4): 1

-- Right-Aligned Increasing Triangle --
Enter number of rows (n): 4
      *
     **
    ***
   ****

===== RIGHT-ALIGNED PATTERNS MENU =====
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter choice (1-4): 2

-- Right-Aligned Increasing Number Triangle --
Enter number of rows (n): 4
      1
     1 2
    1 2 3
   1 2 3 4

===== RIGHT-ALIGNED PATTERNS MENU =====
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter choice (1-4): 3

-- Right-Aligned Decreasing Triangle --
Enter number of rows (n): 4
   ***
  **
 *
 

===== RIGHT-ALIGNED PATTERNS MENU =====
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter choice (1-4): 4
Exiting program ...

(kali@kali)~/HDOOC_Day26$
```